

Week 12-Random Effects

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1 Random effects (Chapter 10, Faraway)

We illustrate the fitting methods using some data from an experiment to test the paper brightness depending on a shift operator described in Sheldon (1960). The pulp data frame has 20 rows and 2 columns. Data comes from an experiment to test the paper brightness depending on a shift operator.

```
data(pulp, package="faraway")
head(pulp)
```

```
##    bright operator
## 1    59.8        a
## 2    60.0        a
## 3    60.8        a
## 4    60.8        a
## 5    59.8        a
## 6    59.8        b
```

1.1 Estimation

1.1.1 ANOVA estimators

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i \in \{1, 2, 3, 4\}, \quad j = 1, \dots, 5$$

$$E(\alpha_i) = E(\varepsilon_{ij}) = 0, \quad \text{Var}(\alpha_i) = \sigma_\alpha^2, \quad \text{Var}(\varepsilon_{ij}) = \sigma^2$$

```
op <- options(contrasts=c("contr.sum", "contr.poly"))
lmod <- aov(bright ~ operator, pulp)
summary(lmod)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## operator      3   1.34   0.4467   4.204 0.0226 *
## Residuals    16   1.70   0.1062
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

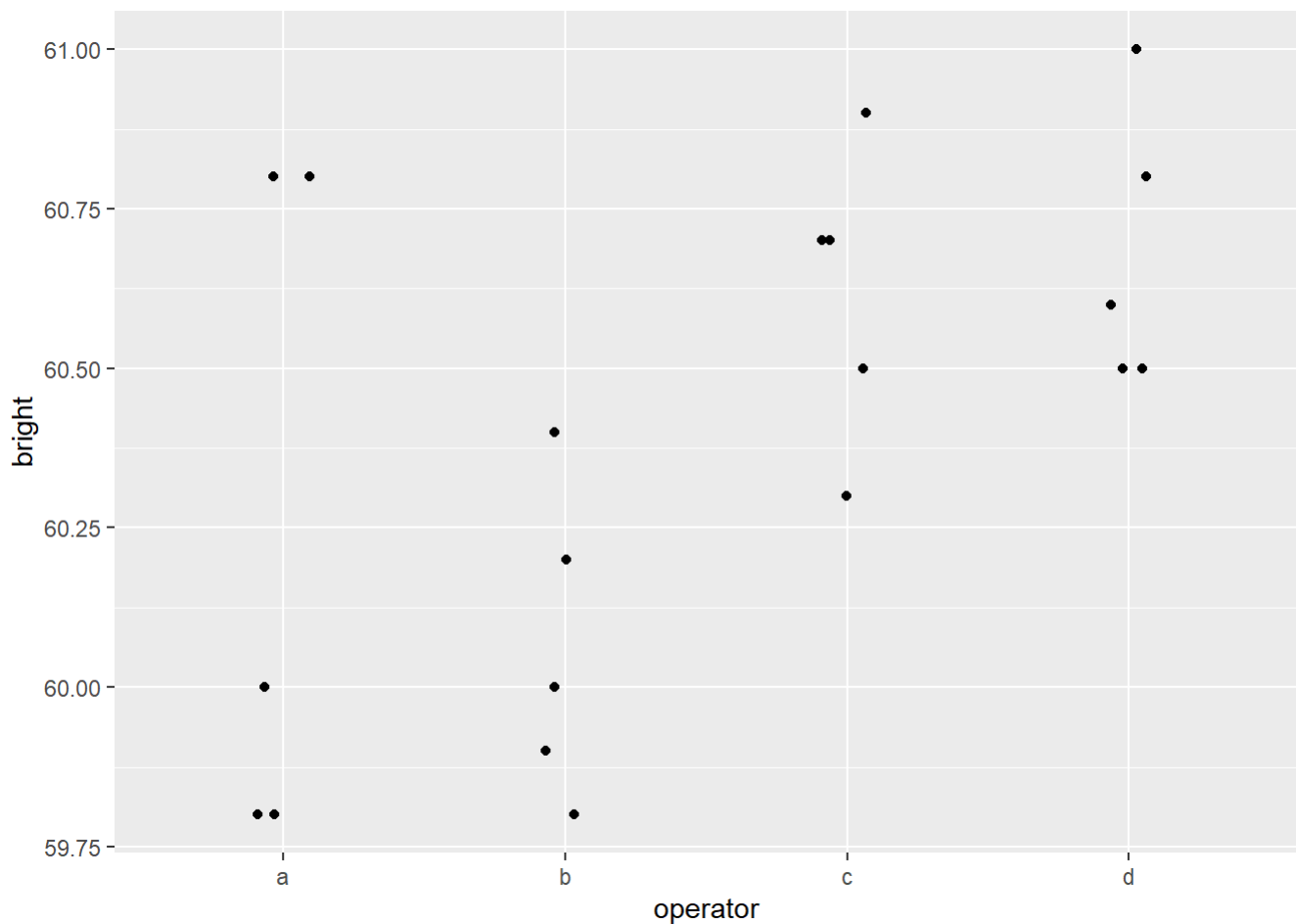
$$\hat{\sigma}^2 = 0.106$$

```
(0.447-0.106)/5
```

```
## [1] 0.0682
```

$$\hat{\sigma}_\alpha^2 = \frac{MSA - MSE}{5} = \frac{0.447 - 0.106}{5} = 0.068$$

```
library(ggplot2)
ggplot(pulp, aes(x=operator, y=bright))+geom_point(position = position_jitter(width=0.1, height=0.0))
```



```
coef(lmod)
```

## (Intercept)	operator1	operator2	operator3
## 60.40	-0.16	-0.34	0.22

```
options(op)
```

$$\hat{\mu} = 60.40 = \bar{y}_{..}$$

$$\sum_{i=1}^4 \alpha_i = 0$$

$$(\hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3, \hat{\alpha}_4) = (-0.16, -0.34, 0.22, 0.16 + 0.34 - 0.22 = 0.28)$$

1.1.2 Restricted Maximum Likelihood (REML) estimators

R package for fitting mixed effects models:

-nlme: original R package (Pinheiro and Bates, 2000)

-lme4: generally more capable especially for larger datasets. (Bates, 2005)

The original R package for fitting mixed effects models was nlme as described in Pinheiro and Bates (2000). Subsequently Bates (2005) introduced the package lme4.

The default fitting method for lme4 is REML.

```
library(lme4)
```

```
## Loading required package: Matrix
```

```
mmod <- lmer(bright ~ 1+(1|operator), pulp)
summary(mmod)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: bright ~ 1 + (1 | operator)
## Data: pulp
##
## REML criterion at convergence: 18.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.4666 -0.7595 -0.1244  0.6281  1.6012
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## operator (Intercept) 0.06808  0.2609
## Residual              0.10625  0.3260
## Number of obs: 20, groups: operator, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  60.4000    0.1494   404.2
```

-Fixed effect: just the intercept

-Random effect: the data is grouped by operator and the random effect is constant within each group.

This is a balanced design, and $\hat{\sigma}^2 = 0.106$, $\hat{\sigma}_\alpha^2 = 0.068$, $\hat{\mu} = 60.4$ are identical to the ANOVA method.

1.1.3 Maximum Likelihood (ML) estimators

```
smmod <- lmer(bright ~ 1+(1|operator), pulp, REML=FALSE)
summary(smmod)
```

```
## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: bright ~ 1 + (1 | operator)
## Data: pulp
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    22.5    25.5     -8.3     16.5      17
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.50554 -0.78116 -0.06353  0.65850  1.56232
##
## Random effects:
## Groups Name Variance Std.Dev.
## operator (Intercept) 0.04575 0.2139
## Residual 0.10625 0.3260
## Number of obs: 20, groups: operator, 4
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 60.4000 0.1294 466.7
```

The between-subjects SD, 0.21, is smaller than with the REML method (0.26) as the ML method biases the estimates towards zero. The fixed effects are unchanged.

1.2 Inference

1.2.1 LR test

$$H_0 : \text{no group effect, } y_{ij} = \mu + \epsilon_{ij}$$

$$H_1 : y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

Parameters:

$$-H_0 : \mu, \sigma^2$$

$$-H_1 : \mu, \sigma^2, \sigma_\alpha^2$$

```
nullmod <- lm(bright ~ 1, pulp)
```

```
lrtstat <- as.numeric(2*(logLik(smod)-logLik(nullmod)))
pvalue <- pchisq(lrtstat,1,lower=FALSE)
data.frame(lrtstat, pvalue)
```

```
## lrtstat pvalue
## 1 2.568371 0.1090199
```

The p-value is now well above the 5% significance level. We cannot say that this result is necessarily wrong, but the use of the χ^2 approximation does cause us to doubt the result.

1.2.2 Parametric Bootstrap

Under H_0 , $y \sim N(\mu, \sigma^2)$. Use the estimates from nullmod.

The simulate function makes it simple to generate a sample from a model:

```
y <- simulate(nullmod)
```

Taking the data we generate, we fit both the null and alternative models and then compute the LRT. We repeat the process 1000 times:

```
lrstat <- numeric(1000)
set.seed(123)
for(i in 1:1000){
  y <- unlist(simulate(nullmod))
  bnull <- lm(y ~ 1)
  balt <- lmer(y ~ 1 + (1|operator), pulp, REML=FALSE)
  lrstat[i] <- as.numeric(2*(logLik(balt)-logLik(bnull)))
}
```

$(lr^{(1)}, lr^{(2)}, \dots, lr^{(1000)})$: the LR test statistic under H_0

$$p\text{-value} = \frac{\#\{k; lr > lr^{(k)}\}}{1000}$$

```
mean(lrstat > 2.5684)
```

```
## [1] 0.019
```

The standard error for this estimate:

```
sqrt(0.019*0.981/1000)
```

```
## [1] 0.004317291
```

So we can be fairly sure the p-value < 0.05. If in doubt, do some more replications to make sure.

The RLRsim package of Scheipl et al. (2008) can be used to test random effect terms:

```
#install.packages("RLRsim")
library(RLRsim)
exactLRT(smod, nullmod)
```

```
## No restrictions on fixed effects. REML-based inference preferable.
```

```
##
## simulated finite sample distribution of LRT. (p-value based on 10000
## simulated values)
##
## data:
## LRT = 2.5684, p-value = 0.0241
```

The result is obtained with less computing time than our explicitly worked example. The difference in the outcomes is within the sampling error.

The parametric bootstrap can also be used to construct confidence intervals for the parameters.

Assume we want to find the 95% CI for σ_{α}^2 .

```
VarCorr(mmod)
```

```
## Groups   Name      Std.Dev.  
## operator (Intercept) 0.26093  
## Residual              0.32596
```

```
as.data.frame(VarCorr(mmod))
```

```
##      grp      var1 var2      vcov      sdcor  
## 1 operator (Intercept) <NA> 0.06808334 0.2609278  
## 2 Residual              <NA> <NA> 0.10625000 0.3259601
```

```
bsd <- numeric(1000)  
for(i in 1:1000){  
  y <- unlist(simulate(mmod))  
  bmod <- refit(mmod, y)  
  bsd[i] <- as.data.frame(VarCorr(bmod))$sdcor[1]  
}
```

```
quantile(bsd, c(0.025, 0.975))
```

```
##      2.5%      97.5%  
## 0.0000000 0.5089133
```

```
confint(mmod, method="boot")
```

```
##      2.5 %      97.5 %  
## .sig01      0.00000 0.5075998  
## .sigma      0.21519 0.4400637  
## (Intercept) 60.11339 60.6808626
```